MULTIMEDIA UNIVERSITY

TPC6323 – Parallel Computing
Trimester March/April 2026 (2610)

Project (40%)

Overview:
This project objective is to convey parallel computing ideas and solutions though effective interaction with diverse stakeholder (CLO3).

In this project, you will develop a parallel computing application based on the recent research and solve the real-world problems using C# parallel programming language. **The chosen title CANNOT be the same as the FYP or any other subject's project title.** Refer the rubrics of parallel computing application.

No	Categories	Example Topics in C# Parallelism
1	Signal Processing	Parallel Graph Processing, Parallel Signal Processing, Parallel Data Compression and Parallel Image Segmentation
2	Cryptography	Parallel Symmetric Encryption (AES), Parallel Asymmetric Encryption (RSA), Parallel Hash Functions, Parallel Digital Signatures and Parallel Key Exchange
3	Machine Learning	Parallel Random Forest Training, Parallel K-Means Clustering and Parallel Deep Learning
4	Computer Graphic Programming	Parallel Ray Tracing, Parallel Rasterization and Parallel Volume Rendering
5	Optimization Algorithms	Parallel Particle Swarm Optimization, Parallel Sequential Minimal Optimization, Parallel Quadratic Programming, Parallel Energy Valley Optimizer and Parallel Spider Wasp Optimizer

Instructions:

1. This is a group project with a maximum of **4 members** in a group.

2. The act of copying codes from others, and then cosmetically changing the codes so that they appear different will result in ZERO marks for both groups.

3. Please specify your group members in Microsoft Team Parallel Computing. The first name in the group will be the project leader.

4. The **project leader** is required to **upload the softcopy of the following files in eBWise**:

   a) Report (follow the Report Template). The similarity index of the report must be less than 10%. AI writing index of the report must be less than 20%. Marks will be deducted from the report if the similarity percentage exceeds 10% or AI writing index percentage exceeds 20%.
   b) Prototype for the proposed parallel computing application.
   c) Every group is required to present and demonstrate the proposed parallel computing.
   d) Group project contribution form. Non-contributing members may receive zero marks.

5. The last date to upload your work is **21 June 2026 (Sunday) 23:59**. 10 marks will be deducted for each overdue day.

Report Template

Parallel Computing Title

Firstname Lastname [1,*], Firstname Lastname [1], Firstname Lastname[1] and Firstname Lastname [1,*]

[1] Faculty of Information Science and Technology, Multimedia University, 75450 Melaka, Malaysia * Correspondence: e-mail@e-mail.com (project leader's email)

Abstract: A single paragraph of about 200 words maximum. For research articles, abstracts should give a pertinent overview of the work.
Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article yet reasonably common within the subject discipline.)

---

1. Introduction

Provide an overview of the parallel computing domains. Discuss the limitations of existing expert systems in a similar domain and offer an overview of the proposed parallel computing applications. In-text citation follows APA format.

2. Research Background

Identify and describe at least 15 related parallel computing in the same domain. Explain their purpose, features, technical details of the methodology or algorithm used, and their strengths and limitations in a table. As well as critical thinking with strong justifications.

3. Methodology

Detail the programming languages or tools used, provide graphical or tabular representation of the rules (specify the source reference of rules), describe the main rules, including code snippets and execution screenshots, and highlight any special features. This section may be divided by subheadings.

Table 1. This is a table. Tables should be placed in the main text near to the first time they are cited.

Title 1	Title 2	Title 3
entry 1	data	data
entry 2	data	data [1]

[1] Tables may have a footer.

4. Results

Provide the experimental results and comparison of the recent 5 parallel computing algorithms, as well as with the existing works. May include diagrams.

5. Conclusion

Concluding remarks of the project (what expert system has been proposed, using what algorithms/methods, etc). Future research directions may also be highlighted.

References

References must be numbered in order of appearance in the text (including citations in tables and legends) and listed individually at the end of the manuscript. We recommend preparing the references with a bibliography software package, such as EndNote, ReferenceManager or Zotero to avoid typing mistakes and duplicated references. Include the digital object identifier (DOI) for all references where available.

1. Author 1, A.B.; Author 2, C.D. Title of the article. *Abbreviated Journal Name* **Year**, *Volume*, page range.
2. Author 1, A.; Author 2, B. Title of the chapter. In *Book Title*, 2nd ed.; Editor 1, A., Editor 2, B., Eds.; Publisher: Publisher Location, Country, 2007; Volume 3, pp. 154–196.
3. Author 1, A.; Author 2, B. *Book Title*, 3rd ed.; Publisher: Publisher Location, Country, 2008; pp. 154–196.
4. Author 1, A.B.; Author 2, C. Title of Unpublished Work. *Abbreviated Journal Name* year, *phrase indicating stage of publication (submitted; accepted; in press)*.
5. Author 1, A.B. (University, City, State, Country); Author 2, C. (Institute, City, State, Country). Personal communication, 2012.
6. Author 1, A.B.; Author 2, C.D.; Author 3, E.F. Title of Presentation. In Proceedings of the Name of the Conference, Location of Conference, Country, Date of Conference (Day Month Year).
7. Author 1, A.B. Title of Thesis. Level of Thesis, Degree-Granting University, Location of University, Date of Completion.
8. Title of Site. Available online: URL (accessed on Day Month Year).

Documentation Assessment Rubrics (40%)

Item	Missing or Unacceptable (0-4)	Poor (5-9)	Accomplished (10-15)	Good (16-20)
Introduction	The proposed project either already exists or differs only slightly from an existing one.	Small adjustments have been made to ideas derived from existing systems.	Ideas adapted from existing systems, enhanced with additional creative elements.	Most of the ideas are innovative and original.
Research Background	The background study is taken directly from the literature without paraphrasing.	The background study is lengthy, with content extracted directly from the literature and lacking critical evaluation.	The background study is concise and well-organized, integrating critical and logical insights from peer-reviewed theoretical and research literature.	The background study is clear and concise, effectively incorporating critical and logical details from peer-reviewed theoretical and research sources.
Methodology	A brief outline of the proposed method is provided but lacks clarity, explanation, or relevance.	An introduction to the application of the methods is included but is missing examples, clear understanding, or adequate explanation.	The system design is well-illustrated and accompanied by clear, detailed explanations.	The system design is well-illustrated and supported by thorough and effective explanations.
Results	The testing methods were absent or poorly aligned with the data and research design, leading to confusing results.	Testing methods were identified; however, the results were unclear, incomplete, or lacked relevance to the research questions, data, or research design.	Testing methods were appropriately identified, and the results were presented clearly. All aspects were relevant to the research question and design, with sufficient metrics or measurements applied.	Testing methods and result presentations were highly detailed, specific, clear, well-structured, and appropriately aligned with the research questions and design. Additional metrics or measurements were also employed.
Conclusion	The conclusion is either absent or entirely fails to address the topic.	The conclusion exists but is incomplete or underdeveloped.	The conclusion effectively summarizes the key points of the work.	The conclusion is comprehensive, concise, and highly effective.

Prototype Assessment Rubrics (60%)

Item	Poor (0-4)	Accomplished (5-7)	Good (8-10)
User interface / Output (10%)	Most of the information/outputs generated are inaccurate, and the layout is disorganized.	The information/outputs generated are mostly accurate but contain some errors, with an organized layout.	The information/outputs generated are entirely accurate, and the layout is well-structured and highly organized.
Programming (15%)	The final product contains numerous logical errors, with many actions lacking proper exception handling. Solutions are overly simplistic, indicating a need for improvement in programming skills.	Majority of the work is logical, certain steps to achieve specific tasks are overly tedious or unnecessarily complex. The program algorithm displays an acceptable level of complexity.	The program demonstrates a correct and logical flow with well-handled exceptions. It showcases a strong or advanced level of algorithmic complexity and programming proficiency.
Degree of completion (10%)	A significant amount of work remains incomplete, and the basic requirements have not been met.	The interface includes all necessary features within the required scope; however, some are simplified, or one or two features are absent.	The interface incorporates all required features, meeting or even exceeding the scope of the requirements.
System implementation (15%)	The final product was developed using a different system design or approach that is unrelated to the original proposal.	The final product aligns with most aspects of the system design; however, certain elements differ from the original specifications.	The final product completely adheres to the proposed system design in every aspect.
Presentation and on-the-spot coding (10%)	Lacks clarity about the work produced and may even struggle to locate the source code.	Aware of where the source code is but may not fully understand the reasoning behind certain aspects of the work.	Thorough understanding of every aspect of the work completed.